# 实验 3　IPSec VPN

## 一、 实验目的

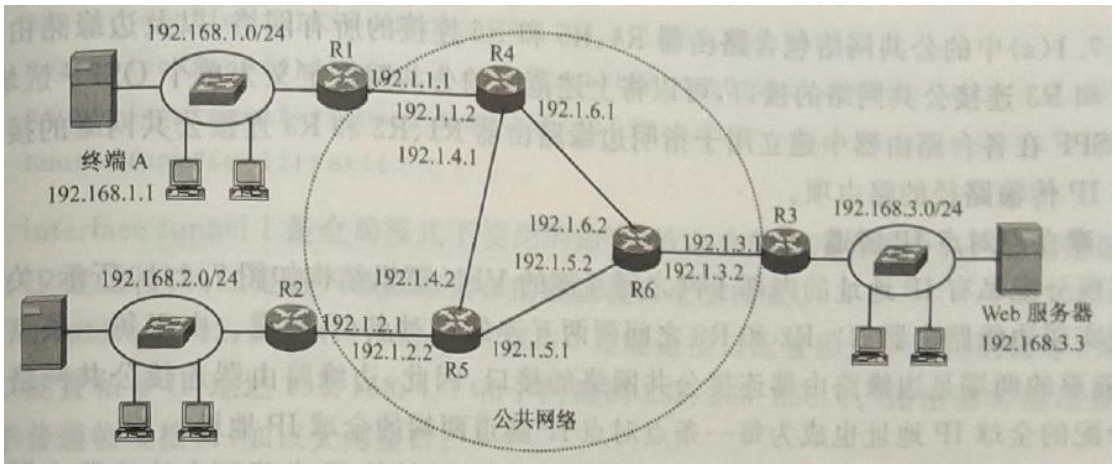
1. 掌握 ISAKMP 策略配置过程
2. 掌握 IPSec 参数配置过程
3. 掌握 IPSec 安全关联建立过程
4. 验证封装安全净荷 ESP 报文的封装过程。
5. 验证基于 IPSec VPN 的数据传输过程

## 二、 实验环境

1. Cisco Packet Tracer 软件作为实验平台

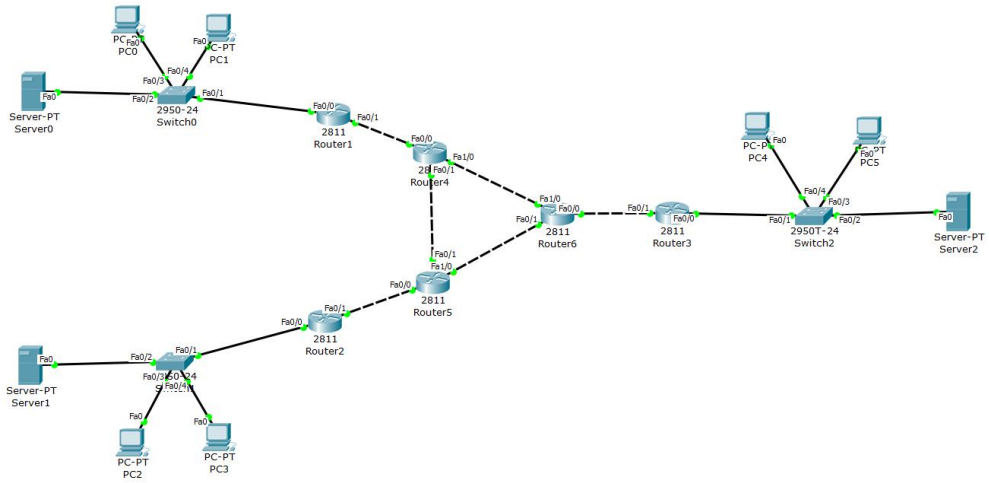
## 三、 实验内容

1. 参考《实验 3-1 附录》，使用 Cisco Packet Tracer 平台完成网络结构基本配置。
2. 实现《实验 3-2 附录》IPSec VPN 参数配置、安全关联建立和数据传输过程。




### 3.1

首先完成 3-1 的实验部分
首先搭建完毕如图所示的实验拓扑图。




首先配置 router1 的信息，由于指令过长，在此仅贴出日志输出，之后的 Ipsec 实验贴出 Ipsec 配置的截图。

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router1
Router1(config)#interface FastEthernet0/0
Router1(config-if)#no shutdown

Router1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router1(config-if)#ip address 192.168.1.254 255.255.255.0
Router1(config-if)#exit
Router1(config)#interface fe0/1
^
% Invalid input detected at '^' marker.
Router1(config)#interface f0/1
Router1(config-if)#no shutdown

Router1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router1(config-if)#ip address 192.1.1.1 255.255.255.0
Router1(config-if)#exit
Router1(config)#router ospf 01
Router1(config-router)#network 192.1.1.0 0.0.0.255 area1
^
% Invalid input detected at '^' marker.
```

```
Router1(config-router)#network 192.1.1.0 0.0.0.255 area1
^
% Invalid input detected at '^' marker.
Router1(config-router)#network 192.1.1.0 0.0.0.255 area 1
Router1(config-router)#exit
Router1(config)#int tunnel 1

Router1(config-if)#
%LINK-5-CHANGED: Interface Tunnel1, changed state to up

Router1(config-if)#ip address 192.168.4.1 255.255.255.0
Router1(config-if)#tunnel source fa0/1
Router1(config-if)#tunnel destination 192.1.2.1
Router1(config-if)#exit
Router1(config)#int tunnel 2

Router1(config-if)#
%LINK-5-CHANGED: Interface Tunnel2, changed state to up

Router1(config-if)#ip address 192.168.5.1 255.255.255.0
Router1(config-if)#tunnel source fa0/1
Router1(config-if)#tunnel destination 192.1.3.1
Router1(config-if)#exit
Router1(config)#router rip
Router1(config-router)#network 192.168.1.0
Router1(config-router)#network 192.168.4.0
Router1(config-router)#network 192.168.5.0
Router1(config-router)#exit
Router1(config)#
Router#
```

之后配置 router2 的信息

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.    End with CNTL/Z.
Router(config)#hostname Router2
Router2(config)#int fa0/0
Router2(config-if)#no shutdown

Router2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router2(config-if)#ip addr 192.168.2.254 255.255.255.0
```

```
Router2(config-if)#exit
Router2(config)#int fa0/1
Router2(config-if)#no shutdown

Router2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router2(config-if)#ip addr 192.1.2.1 255.255.255.0
Router2(config-if)#exit
Router2(config)#router ospf 02
Router2(config-router)#network 192.1.2.0 0.0.0.255 area 1
Router2(config-router)#exit
Router2(config)#int tunnel 1

Router2(config-if)#
%LINK-5-CHANGED: Interface Tunnel1, changed state to up

Router2(config-if)#ip addr 192.168.4.2 255.255.255.0
Router2(config-if)#tunnel source fa0/1
Router2(config-if)#tunnel destination 192.1.1.1
Router2(config-if)#exit
Router2(config)#interface tunnel 3

Router2(config-if)#
%LINK-5-CHANGED: Interface Tunnel3, changed state to up

Router2(config-if)#ip addr 192.168.6.1 255.255.255.0
Router2(config-if)#tunnel source fa0/1
Router2(config-if)#tunnel dest 192.1.3.1
Router2(config-if)#exit
Router2(config)#router rip
Router2(config-router)#network 192.168.2.0
Router2(config-router)#network 192.168.4.0
Router2(config-router)#network 192.168.6.0
Router2(config-router)#exit
Router2(config)#
```

之后配置 router3 的信息

```
Router>enable
Router#conf ter
Enter configuration commands, one per line.    End with CNTL/Z.
Router(config)#hostname Router3
Router3(config)#int fa0/0
Router3(config-if)#no shutdown
```

```
Router3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router3(config-if)#ip addr 192.168.3.254 255.255.255.0
Router3(config-if)#exit
Router3(config)#int fa0/1
Router3(config-if)#no shutdown

Router3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router3(config-if)#ip addr 192.1.3.1 255.255.255.0
Router3(config-if)#exit
Router3(config)#router ospf 03
Router3(config-router)#network 192.1.3.0 0.0.0.255 area 1
Router3(config-router)#exit
Router3(config)#int tunnel 2

Router3(config-if)#
%LINK-5-CHANGED: Interface Tunnel2, changed state to up

Router3(config-if)#ip addr 192.168.5.2 255.255.255.0
Router3(config-if)#tunnel source fa0/1
Router3(config-if)#tunnel dest 192.1.1.1
Router3(config-if)#exit
Router3(config)#int tunnel 3

Router3(config-if)#
%LINK-5-CHANGED: Interface Tunnel3, changed state to up

Router3(config-if)#ip addr 192.168.6.2 255.255.255.0
Router3(config-if)#tunnel source fa0/1
Router3(config-if)#tunnel dest 192.1.2.1
Router3(config-if)#exit
Router3(config)#router rip
Router3(config-router)#network 192.168.3.0
Router3(config-router)#network 192.168.5.0
Router3(config-router)#network 192.168.6.0
Router3(config-router)#exit
Router3(config)#
```

之后来配置 router4 的信息

```
Router>enable
```

```
Router#conf ter
Enter configuration commands, one per line.    End with CNTL/Z.
Router(config)#hostname Router 4
                                          ^
% Invalid input detected at '^' marker.

Router(config)#hostname Router4
Router4(config)#int fa0/0
Router4(config-if)#no shutdown

Router4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router4(config-if)#ip addr    192.1.1.2 255.255.255.0
Router4(config-if)#exit
Router4(config)#int fa0/1
Router4(config-if)#no shutdown

Router4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router4(config-if)#ip addr 192.1.4.1 255.255.255.0
Router4(config-if)#exit
Router4(config)#int fa1/0
Router4(config-if)#no shutdown

Router4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router4(config-if)#ip addr 192.1.6.1 255.255.255.0
Router4(config-if)#exit
Router4(config)#router ospf 04
Router4(config-router)#network 192.1.1.0 0.0.0.255 area 1
Router4(config-router)#network 192.1.4.0 0.0.0.255 area 1
Router4(config-router)#network 1
00:27:22: %OSPF-5-ADJCHG: Process 4, Nbr 192.168.1.254 on FastEthernet0/0 from LOADING to
FULL, Load
Router4(config-router)#network 192.1.6.0 0.0.0.255 area 1
Router4(config-router)#exit
Router4(config)#
```

之后来配置 Router5 的信息

```
Router>enable
```

```
Router#hostname Router5
                        ^
% Invalid input detected at '^' marker.

Router#conf ter
Enter configuration commands, one per line.    End with CNTL/Z.
Router(config)#hostaname Router5
                            ^
% Invalid input detected at '^' marker.

Router(config)#hostname Router5
Router5(config)#int fa0.0
                                  ^
% Invalid input detected at '^' marker.

Router5(config)#int fa0/0
Router5(config-if)#no shutdown

Router5(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router5(config-if)#ip addr 192.1.2.2 255.255.255.0
Router5(config-if)#exit
Router5(config)#int fa0/1
Router5(config-if)#no shutdown

Router5(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router5(config-if)#ip addr 192.1.4.2 255.255.255.0
Router5(config-if)#exit
Router5(config)#int fa1/0
Router5(config-if)#no shutdown

Router5(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router5(config-if)#ip addr 192.1.5.1 255.255.255.0
Router5(config-if)#exit
Router5(config)#router ospf 05
```

```
Router5(config-router)#network 192.1.2.0 0.0.0.255 area 1
Router5(config-router)#network 192.1.4.0 0.0.0.255
00:30:36: %OSPF-5-ADJCHG: Process 5, Nbr 192.168.2.254 on FastEthernet0/0 from LOADING to
FULL, Loading Done

% Incomplete command.
Router5(config-router)#network 192.1.4.0 0.0.0.255 area 1
Router5(config-router)#network 192.1.5.0 0.0.0.255 area 1
Router5(config-router)#exit
00:30:59: %OSPF-5-ADJCHG: Process 5, Nbr 192.1.6.1 on FastEthernet0/1 from LOADING to FULL,
Loading Done

Router5(config)#
```

之后来配置 router6 的信息

```
Router>enable
Router#conf ter
Enter configuration commands, one per line.    End with CNTL/Z.
Router(config)#hostname Router6
Router6(config)#int fa0/0
Router6(config-if)#no shutdown

Router6(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router6(config-if)#ip addr 192.1.3.2 255.255.255.0
Router6(config-if)#exit
Router6(config)#int fa0/1
Router6(config-if)#no shutdown

Router6(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router6(config-if)#ip addr 192.1.5.2 255.255.255.0
Router6(config-if)#exit
Router6(config)#int fa1/0
Router6(config-if)#no shutdown

Router6(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
```

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router6(config-if)#ip addr 192.1.6.2 255.255.255.0
Router6(config-if)#exit
Router6(config)#router ospf 06
Router6(config-router)#network 192.1.3.0 0.0.0.255 area 1
Router6(config-router)#network 192.1.5.0 0.0.0.255 area 1
Router6(config-router)#network 192.1.6.0 0.0.0.255 area 1
Router6(config-router)#exit
00:34:00: %OSPF-5-ADJCHG: Process 6, Nbr 192.168.3.254 on FastEthernet0/0 from LOADING to FULL, Loading Done

Router6(config)#
00:34:08: %OSPF-5-ADJCHG: Process 6, Nbr 192.1.5.1 on FastEthernet0/1 from LOADING to FULL, Loading Done

00:34:11: %OSPF-5-ADJCHG: Process 6, Nbr 192.1.6.1 on FastEthernet1/0 from LOADING to FULL, Loading Done

之后查看每个路由器的路由表
Router1

Routing Table for Router1

| Type | Network | Port | Next Hop IP | Metric |
|---|---|---|---|---|
| C | 192.1.1.0/24 | FastEthernet0/1 | --- | 0/0 |
| O | 192.1.2.0/24 | FastEthernet0/1 | 192.1.1.2 | 110/3 |
| O | 192.1.3.0/24 | FastEthernet0/1 | 192.1.1.2 | 110/3 |
| O | 192.1.4.0/24 | FastEthernet0/1 | 192.1.1.2 | 110/2 |
| O | 192.1.5.0/24 | FastEthernet0/1 | 192.1.1.2 | 110/3 |
| O | 192.1.6.0/24 | FastEthernet0/1 | 192.1.1.2 | 110/2 |
| C | 192.168.1.0/24 | FastEthernet0/0 | --- | 0/0 |
| R | 192.168.2.0/24 | Tunnel1 | 192.168.4.2 | 120/1 |
| R | 192.168.3.0/24 | Tunnel2 | 192.168.5.2 | 120/1 |
| C | 192.168.4.0/24 | Tunnel1 | --- | 0/0 |
| C | 192.168.5.0/24 | Tunnel2 | --- | 0/0 |
| R | 192.168.6.0/24 | Tunnel2 | 192.168.5.2 | 120/1 |
| R | 192.168.6.0/24 | Tunnel1 | 192.168.4.2 | 120/1 |

Router2

Routing Table for Router2

| Type | Network | Port | Next Hop IP | Metric |
|---|---|---|---|---|
| O | 192.1.1.0/24 | FastEthernet0/1 | 192.1.2.2 | 110/3 |
| C | 192.1.2.0/24 | FastEthernet0/1 | --- | 0/0 |
| O | 192.1.3.0/24 | FastEthernet0/1 | 192.1.2.2 | 110/3 |
| O | 192.1.4.0/24 | FastEthernet0/1 | 192.1.2.2 | 110/2 |
| O | 192.1.5.0/24 | FastEthernet0/1 | 192.1.2.2 | 110/2 |
| O | 192.1.6.0/24 | FastEthernet0/1 | 192.1.2.2 | 110/3 |
| R | 192.168.1.0/24 | Tunnel1 | 192.168.4.1 | 120/1 |
| C | 192.168.2.0/24 | FastEthernet0/0 | --- | 0/0 |
| R | 192.168.3.0/24 | Tunnel3 | 192.168.6.2 | 120/1 |
| C | 192.168.4.0/24 | Tunnel1 | --- | 0/0 |
| R | 192.168.5.0/24 | Tunnel3 | 192.168.6.2 | 120/1 |
| R | 192.168.5.0/24 | Tunnel1 | 192.168.4.1 | 120/1 |
| C | 192.168.6.0/24 | Tunnel3 | --- | 0/0 |

Router3

Routing Table for Router3

| Type | Network | Port | Next Hop IP | Metric |
|---|---|---|---|---|
| O | 192.1.1.0/24 | FastEthernet0/1 | 192.1.3.2 | 110/3 |
| O | 192.1.2.0/24 | FastEthernet0/1 | 192.1.3.2 | 110/3 |
| C | 192.1.3.0/24 | FastEthernet0/1 | --- | 0/0 |
| O | 192.1.4.0/24 | FastEthernet0/1 | 192.1.3.2 | 110/3 |
| O | 192.1.5.0/24 | FastEthernet0/1 | 192.1.3.2 | 110/2 |
| O | 192.1.6.0/24 | FastEthernet0/1 | 192.1.3.2 | 110/2 |
| R | 192.168.1.0/24 | Tunnel2 | 192.168.5.1 | 120/1 |
| R | 192.168.2.0/24 | Tunnel3 | 192.168.6.1 | 120/1 |
| C | 192.168.3.0/24 | FastEthernet0/0 | --- | 0/0 |
| R | 192.168.4.0/24 | Tunnel3 | 192.168.6.1 | 120/1 |
| R | 192.168.4.0/24 | Tunnel2 | 192.168.5.1 | 120/1 |
| C | 192.168.5.0/24 | Tunnel2 | --- | 0/0 |
| C | 192.168.6.0/24 | Tunnel3 | --- | 0/0 |

Router4

Routing Table for Router4

| Type | Network | Port | Next Hop IP | Metric |
|---|---|---|---|---|
| C | 192.1.1.0/24 | FastEthernet0/0 | --- | 0/0 |
| O | 192.1.2.0/24 | FastEthernet0/1 | 192.1.4.2 | 110/2 |
| O | 192.1.3.0/24 | FastEthernet1/0 | 192.1.6.2 | 110/2 |
| C | 192.1.4.0/24 | FastEthernet0/1 | --- | 0/0 |
| O | 192.1.5.0/24 | FastEthernet0/1 | 192.1.4.2 | 110/2 |
| O | 192.1.5.0/24 | FastEthernet1/0 | 192.1.6.2 | 110/2 |
| C | 192.1.6.0/24 | FastEthernet1/0 | --- | 0/0 |

Router5

Routing Table for Router5

| Type | Network | Port | Next Hop IP | Metric |
|---|---|---|---|---|
| O | 192.1.1.0/24 | FastEthernet0/1 | 192.1.4.1 | 110/2 |
| C | 192.1.2.0/24 | FastEthernet0/0 | --- | 0/0 |
| O | 192.1.3.0/24 | FastEthernet1/0 | 192.1.5.2 | 110/2 |
| C | 192.1.4.0/24 | FastEthernet0/1 | --- | 0/0 |
| C | 192.1.5.0/24 | FastEthernet1/0 | --- | 0/0 |
| O | 192.1.6.0/24 | FastEthernet0/1 | 192.1.4.1 | 110/2 |
| O | 192.1.6.0/24 | FastEthernet1/0 | 192.1.5.2 | 110/2 |

Router6

Routing Table for Router6

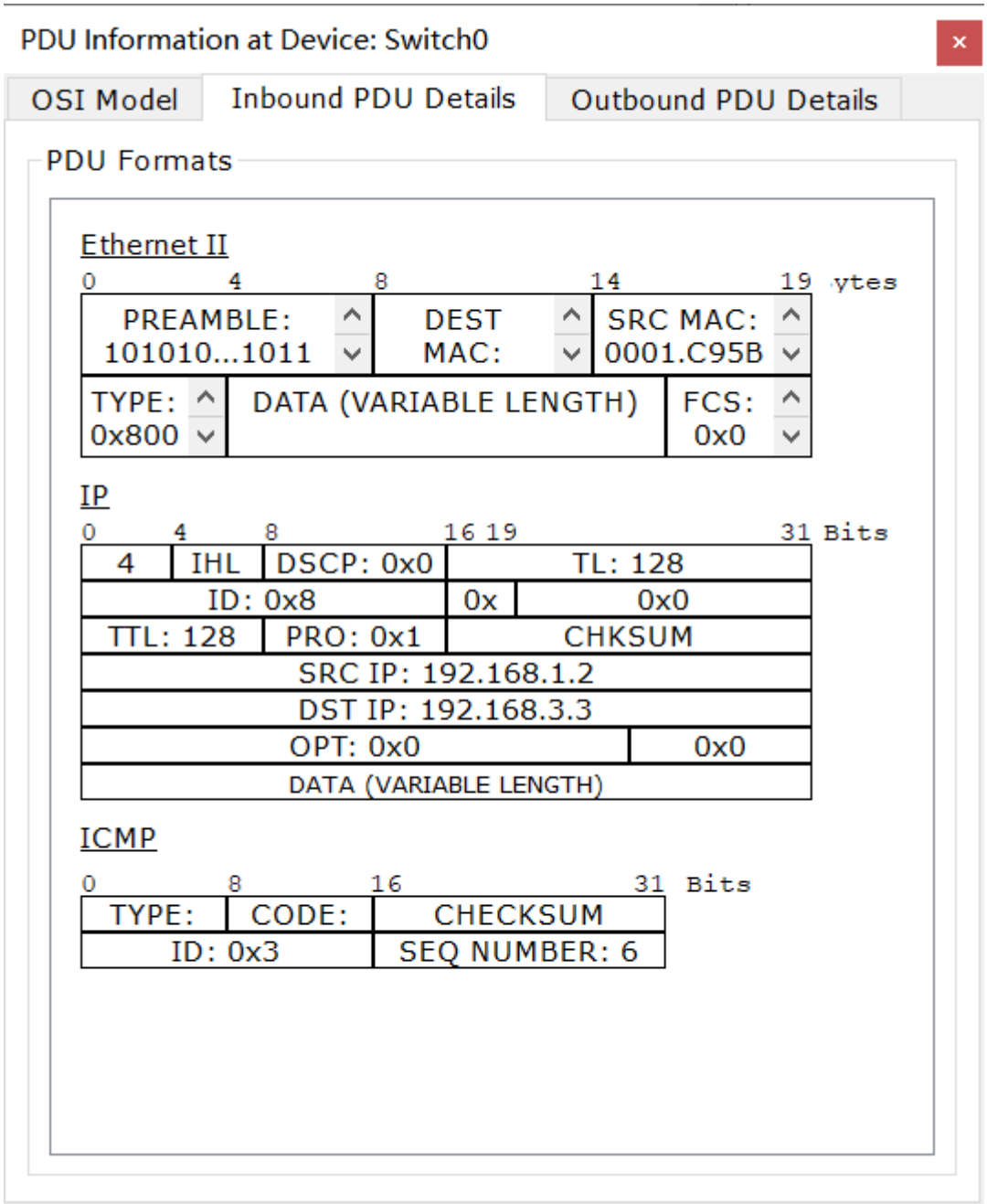
| Type | Network | Port | Next Hop IP | Metric |
|---|---|---|---|---|
| O | 192.1.1.0/24 | FastEthernet1/0 | 192.1.6.1 | 110/2 |
| O | 192.1.2.0/24 | FastEthernet0/1 | 192.1.5.1 | 110/2 |
| C | 192.1.3.0/24 | FastEthernet0/0 | --- | 0/0 |
| O | 192.1.4.0/24 | FastEthernet0/1 | 192.1.5.1 | 110/2 |
| O | 192.1.4.0/24 | FastEthernet1/0 | 192.1.6.1 | 110/2 |
| C | 192.1.5.0/24 | FastEthernet0/1 | --- | 0/0 |
| C | 192.1.6.0/24 | FastEthernet1/0 | --- | 0/0 |

启动 PC0 -> Server2 的 ICMP 传输过程，查看列表显示如下。

Event List

| Vis. | Time(sec) | Last Dev | At Devi | Type | Info |
|---|---|---|---|---|---|
| | 6.002 | PC0 | Switch0 | ICMP | |
| | 6.003 | Switch0 | Router1 | ICMP | |
| | 6.004 | Router1 | Router4 | ICMP | |
| | 6.005 | Router4 | Router6 | ICMP | |
| | 6.006 | Router6 | Router3 | ICMP | |
| | 6.007 | Router3 | Switch2 | ICMP | |
| | 6.008 | Switch2 | Server2 | ICMP | |

内层 ip 数据报格式


PDU Information at Device: Switch0
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
Ethernet II
0 4 8 14 19 ytes
PREAMBLE: 101010...1011
DEST MAC:
SRC MAC: 0001.C95B
TYPE: 0x800
DATA (VARIABLE LENGTH)
FCS: 0x0
IP
0 4 8 16 19 31 Bits
4
IHL
DSCP: 0x0
TL: 128
ID: 0x8
0x
0x0
TTL: 128
PRO: 0x1
CHKSUM
SRC IP: 192.168.1.2
DST IP: 192.168.3.3
OPT: 0x0
0x0
DATA (VARIABLE LENGTH)
ICMP
0 8 16 31 Bits
TYPE:
CODE:
CHECKSUM
ID: 0x3
SEQ NUMBER: 6

外层 ip 数据报格式

PDU Information at Device: Router4

OSI Model | Inbound PDU Details | Outbound PDU Details

PDU Formats

0 4 8 14 19 ytes

| PREAMBLE: 101010...1011 | DEST MAC: | SRC MAC: 0060.2FA9 |
|---|---|---|
| TYPE: 0x800 | DATA (VARIABLE LENGTH) | FCS: 0x0 |

IP

0 4 8 16 19 31 Bits

| | | | | |
|---|---|---|---|---|
| 4 | IHL | DSCP: 0x0 | TL: 20 | |
| ID: 0x1e1 | | | 0x | 0x0 |
| TTL: 255 | | PRO: 0x2f | CHKSUM | |
| SRC IP: 192.1.1.1 | | | | |
| DST IP: 192.1.3.1 | | | | |
| OPT: 0x0 | | | | 0x0 |
| DATA (VARIABLE LENGTH) | | | | |

GRE

0 8 16 31 Bits

| | |
|---|---|
| FLAGS: 0 | PROTOCOL TYPE: 0x800 |

IP

0 4 8 16 19 31 Bits

| | | | | |
|---|---|---|---|---|
| 4 | IHL | DSCP: 0x0 | TL: 128 | |
| ID: 0x8 | | | 0x | 0x0 |
| TTL: 127 | | PRO: 0x1 | CHKSUM | |
| SRC IP: 192.168.1.2 | | | | |
| DST IP: 192.168.3.3 | | | | |
| OPT: 0x0 | | | | 0x0 |
| DATA (VARIABLE LENGTH) | | | | |

ICMP

0 8 16 31 Bits

| | | |
|---|---|---|
| TYPE: | CODE: | CHECKSUM |
| ID: 0x3 | | SEQ NUMBER: 6 |

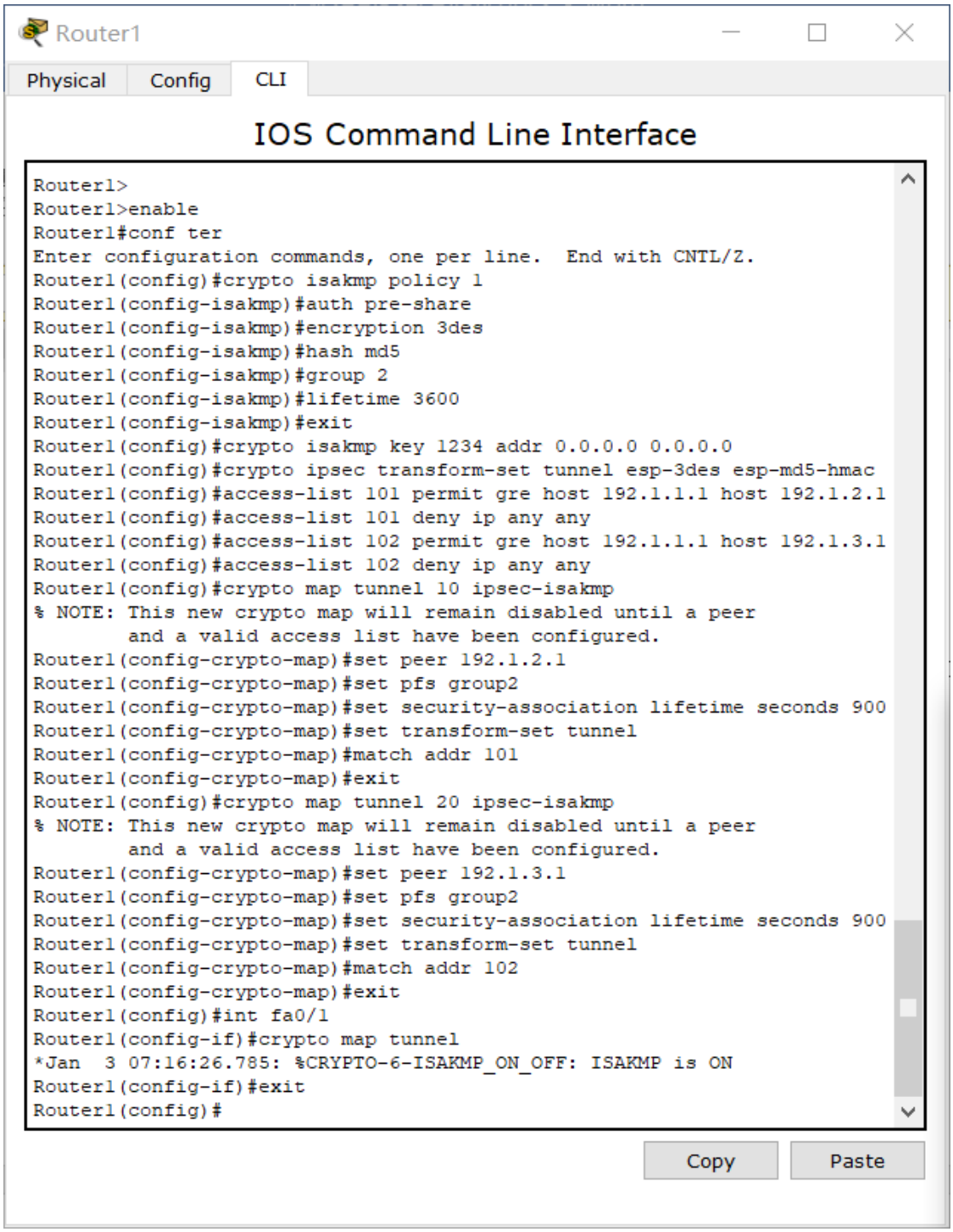
## 3.2

接下来完成 3-2 的实验部分

首先配置 Router1

```
Router1>
Router1>enable
Router1#conf ter
Enter configuration commands, one per line.    End with CNTL/Z.
Router1(config)#crypto isakmp policy 1
Router1(config-isakmp)#auth pre-share
Router1(config-isakmp)#encryption 3des
Router1(config-isakmp)#hash md5
Router1(config-isakmp)#group 2
Router1(config-isakmp)#lifetime 3600
Router1(config-isakmp)#exit
Router1(config)#crypto isakmp key 1234 addr 0.0.0.0 0.0.0.0
Router1(config)#crypto ipsec transform-set tunnel esp-3des esp-md5-hmac
Router1(config)#access-list 101 permit gre host 192.1.1.1 host 192.1.2.1
Router1(config)#access-list 101 deny ip any any
Router1(config)#access-list 102 permit gre host 192.1.1.1 host 192.1.3.1
Router1(config)#access-list 102 deny ip any any
Router1(config)#crypto map tunnel 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router1(config-crypto-map)#set peer 192.1.2.1
Router1(config-crypto-map)#set pfs group2
Router1(config-crypto-map)#set security-association lifetime seconds 900
Router1(config-crypto-map)#set transform-set tunnel
Router1(config-crypto-map)#match addr 101
Router1(config-crypto-map)#exit
Router1(config)#crypto map tunnel 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router1(config-crypto-map)#set peer 192.1.3.1
Router1(config-crypto-map)#set pfs group2
Router1(config-crypto-map)#set security-association lifetime seconds 900
Router1(config-crypto-map)#set transform-set tunnel
Router1(config-crypto-map)#match addr 102
Router1(config-crypto-map)#exit
Router1(config)#int fa0/1
Router1(config-if)#crypto map tunnel
*Jan    3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router1(config-if)#exit
```

Router1(config)#




```
Router1>
Router1>enable
Router1#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#crypto isakmp policy 1
Router1(config-isakmp)#auth pre-share
Router1(config-isakmp)#encryption 3des
Router1(config-isakmp)#hash md5
Router1(config-isakmp)#group 2
Router1(config-isakmp)#lifetime 3600
Router1(config-isakmp)#exit
Router1(config)#crypto isakmp key 1234 addr 0.0.0.0 0.0.0.0
Router1(config)#crypto ipsec transform-set tunnel esp-3des esp-md5-hmac
Router1(config)#access-list 101 permit gre host 192.1.1.1 host 192.1.2.1
Router1(config)#access-list 101 deny ip any any
Router1(config)#access-list 102 permit gre host 192.1.1.1 host 192.1.3.1
Router1(config)#access-list 102 deny ip any any
Router1(config)#crypto map tunnel 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router1(config-crypto-map)#set peer 192.1.2.1
Router1(config-crypto-map)#set pfs group2
Router1(config-crypto-map)#set security-association lifetime seconds 900
Router1(config-crypto-map)#set transform-set tunnel
Router1(config-crypto-map)#match addr 101
Router1(config-crypto-map)#exit
Router1(config)#crypto map tunnel 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router1(config-crypto-map)#set peer 192.1.3.1
Router1(config-crypto-map)#set pfs group2
Router1(config-crypto-map)#set security-association lifetime seconds 900
Router1(config-crypto-map)#set transform-set tunnel
Router1(config-crypto-map)#match addr 102
Router1(config-crypto-map)#exit
Router1(config)#int fa0/1
Router1(config-if)#crypto map tunnel
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router1(config-if)#exit
Router1(config)#
```

之后配置 Router2

Router2>enable

Router2#conf ter

Enter configuration commands, one per line.  End with CNTL/Z.

Router2(config)#crypto isakmp policy 1

Router2(config-isakmp)#auth pre-share

Router2(config-isakmp)#enc 3des

Router2(config-isakmp)#hash md5

```
Router2(config-isakmp)#group 2
Router2(config-isakmp)#lifetime 3600
Router2(config-isakmp)#exit
Router2(config)#crypto isakmp key 1234 addr 0.0.0.0 0.0.0.0
Router2(config)#crypto ipsec transform-set tunnel esp-3des esp-md5-hmac
Router2(config)#al
                        ^
% Invalid input detected at '^' marker.

Router2(config)#acc 101 premit gre host 192.1.2.1 host 192.1.1.1
                                  ^
% Invalid input detected at '^' marker.

Router2(config)#access-list 101 premit gre host 192.1.2.1 host 192.1.1.1
                                             ^
% Invalid input detected at '^' marker.

Router2(config)#acc 101 permit gre host 192.1.2.1 host 192.1.1.1
Router2(config)#acc 101 deny ip any any
Router2(config)#acc 101 permit gre host 192.1.2.1 host 192.1.3.1
Router2(config)#acc 102 permit gre host 192.1.2.1 host 192.1.3.1
Router2(config)#acc 102 deny ip any any
Router2(config)#crypto map tunnel 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router2(config-crypto-map)#set peer 192.1.1.1
Router2(config-crypto-map)#set pfs group2
Router2(config-crypto-map)#set security-association lifetime seconds 900
Router2(config-crypto-map)#set transform-set tunnel
Router2(config-crypto-map)#match addr 101
Router2(config-crypto-map)#exit
Router2(config)#crypto map tunnel 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router2(config-crypto-map)#set peer 192.1.3.1
Router2(config-crypto-map)#set pfs group2
Router2(config-crypto-map)#set security-association lifetime seconds 900
Router2(config-crypto-map)#set transform-set tunnel
Router2(config-crypto-map)#match addr 102
Router2(config-crypto-map)#exit
Router2(config)#int fa0/1
Router2(config-if)#crypto map tunnel
*Jan    3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router2(config-if)#exit
```

Router2(config)#

Router2

Physical Config CLI

IOS Command Line Interface

```
Enter configuration commands, one per line.  End with CNTL/Z.
Router2(config)#crypto isakmp policy 1
Router2(config-isakmp)#auth pre-share
Router2(config-isakmp)#enc 3des
Router2(config-isakmp)#hash md5
Router2(config-isakmp)#group 2
Router2(config-isakmp)#lifetime 3600
Router2(config-isakmp)#exit
Router2(config)#crypto isakmp key 1234 addr 0.0.0.0 0.0.0.0
Router2(config)#crypto ipsec transform-set tunnel esp-3des esp-md5-hmac
Router2(config)#al
                 ^
% Invalid input detected at '^' marker.

Router2(config)#acc 101 premit gre host 192.1.2.1 host 192.1.1.1
                       ^
% Invalid input detected at '^' marker.

Router2(config)#access-list 101 premit gre host 192.1.2.1 host 192.1.1.1
                                ^
% Invalid input detected at '^' marker.

Router2(config)#acc 101 permit gre host 192.1.2.1 host 192.1.1.1
Router2(config)#acc 101 deny ip any any
Router2(config)#acc 101 permit gre host 192.1.2.1 host 192.1.3.1
Router2(config)#acc 102 permit gre host 192.1.2.1 host 192.1.3.1
Router2(config)#acc 102 deny ip any any
Router2(config)#crypto map tunnel 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router2(config-crypto-map)#set peer 192.1.1.1
Router2(config-crypto-map)#set pfs group2
Router2(config-crypto-map)#set security-association lifetime seconds 900
Router2(config-crypto-map)#set transform-set tunnel
Router2(config-crypto-map)#match addr 101
Router2(config-crypto-map)#exit
Router2(config)#crypto map tunnel 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router2(config-crypto-map)#set peer 192.1.3.1
Router2(config-crypto-map)#set pfs group2
Router2(config-crypto-map)#set security-association lifetime seconds 900
Router2(config-crypto-map)#set transform-set tunnel
Router2(config-crypto-map)#match addr 102
Router2(config-crypto-map)#exit
Router2(config)#int fa0/1
Router2(config-if)#crypto map tunnel
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router2(config-if)#exit
Router2(config)#
```

Copy Paste

最后来配置 router3

Router3>

Router3>enable

```
Router3#conf ter
Enter configuration commands, one per line.    End with CNTL/Z.
Router3(config)#auth pre-share
                          ^
% Invalid input detected at '^' marker.

Router3(config)#crypto isakmp policy 1
Router3(config-isakmp)#auth pre-share
Router3(config-isakmp)#enc 3des
Router3(config-isakmp)#hash md5
Router3(config-isakmp)#group 2
Router3(config-isakmp)#lifetime 3600
Router3(config-isakmp)#exit
Router3(config)#crypto isakmp key 1234 addr 0.0.0.0 0.0.0.0
Router3(config)#crypto ipsec transform-set tunnel esp-3des esp-md5-hmac
Router3(config)#acc 101 permit gre host 192.1.3.1 host 192.1.1.1
Router3(config)#acc 101 deny ip any any
Router3(config)#acc 102 permit gre host 192.1.3.1 host 192.1.2.1
Router3(config)#acc 102 deny ip any any
Router3(config)#crypto map tunnel 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router3(config-crypto-map)#set peer 192.1.1.1
Router3(config-crypto-map)#set pfs group2
Router3(config-crypto-map)#set security-association lifetime seconds 900
Router3(config-crypto-map)#set transform-set tunnel
Router3(config-crypto-map)#match addr 101
Router3(config-crypto-map)#exit
Router3(config)#crypto map tunnel 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router3(config-crypto-map)#set peer 192.1.1.1
Router3(config-crypto-map)#set peer 192.1.2.1
Router3(config-crypto-map)#set pfs group2
Router3(config-crypto-map)#set security-association lifetime seconds 900
Router3(config-crypto-map)#set transform-set tunnel
Router3(config-crypto-map)#match addr 102
Router3(config-crypto-map)#exit
Router3(config)#int fa0/1
Router3(config-if)#crypto map tunnel
*Jan    3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router3(config-if)#exit
Router3(config)#
```

Router3

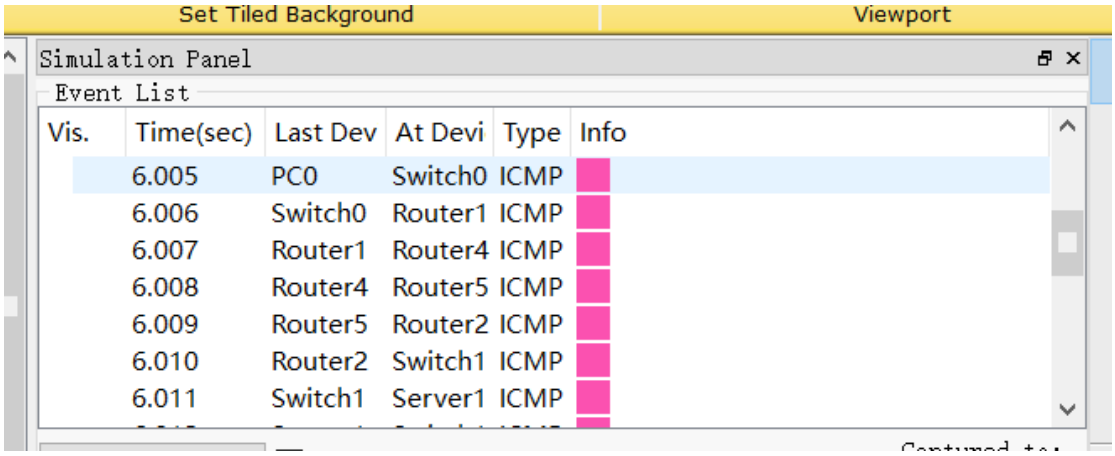
Physical | Config | CLI

IOS Command Line Interface

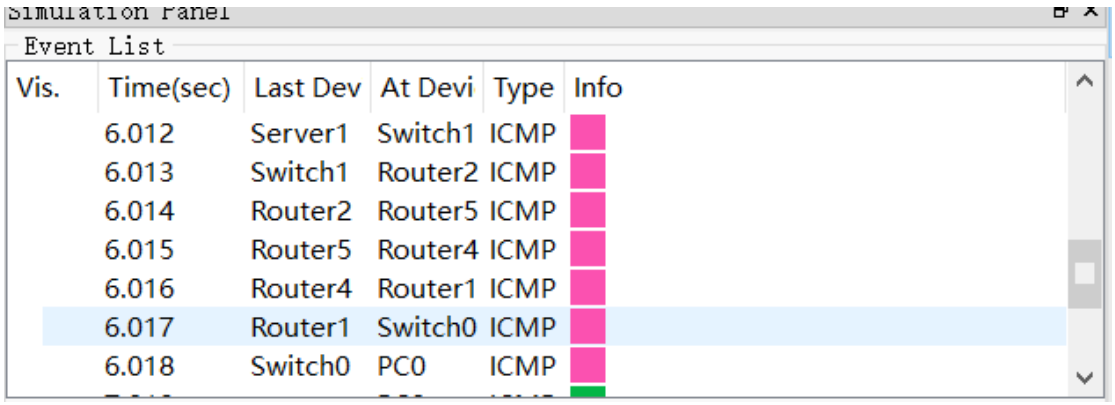
```
Router3>
Router3>enable
Router3#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router3(config)#auth pre-share
                ^
% Invalid input detected at '^' marker.

Router3(config)#crypto isakmp policy 1
Router3(config-isakmp)#auth pre-share
Router3(config-isakmp)#enc 3des
Router3(config-isakmp)#hash md5
Router3(config-isakmp)#group 2
Router3(config-isakmp)#lifetime 3600
Router3(config-isakmp)#exit
Router3(config)#crypto isakmp key 1234 addr 0.0.0.0 0.0.0.0
Router3(config)#crypto ipsec transform-set tunnel esp-3des esp-md5-hmac
Router3(config)#acc 101 permit gre host 192.1.3.1 host 192.1.1.1
Router3(config)#acc 101 deny ip any any
Router3(config)#acc 101 permit gre host 192.1.3.1 host 192.1.2.1
Router3(config)#acc 102 permit gre host 192.1.3.1 host 192.1.2.1
Router3(config)#acc 102 deny ip any any
Router3(config)#crypto map tunnel 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router3(config-crypto-map)#set peer 192.1.1.1
Router3(config-crypto-map)#set pfs group2
Router3(config-crypto-map)#set security-association lifetime seconds 900
Router3(config-crypto-map)#set transform-set tunnel
Router3(config-crypto-map)#match addr 101
Router3(config-crypto-map)#exit
Router3(config)#crypto map tunnel 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router3(config-crypto-map)#set peer 192.1.1.1
Router3(config-crypto-map)#set peer 192.1.2.1
Router3(config-crypto-map)#set pfs group2
Router3(config-crypto-map)#set security-association lifetime seconds 900
Router3(config-crypto-map)#set transform-set tunnel
Router3(config-crypto-map)#match addr 102
Router3(config-crypto-map)#exit
Router3(config)#int fa0/1
Router3(config-if)#crypto map tunnel
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router3(config-if)#exit
Router3(config)#
```

这次启动从 PC0-server1 的 ICMP 传输过程

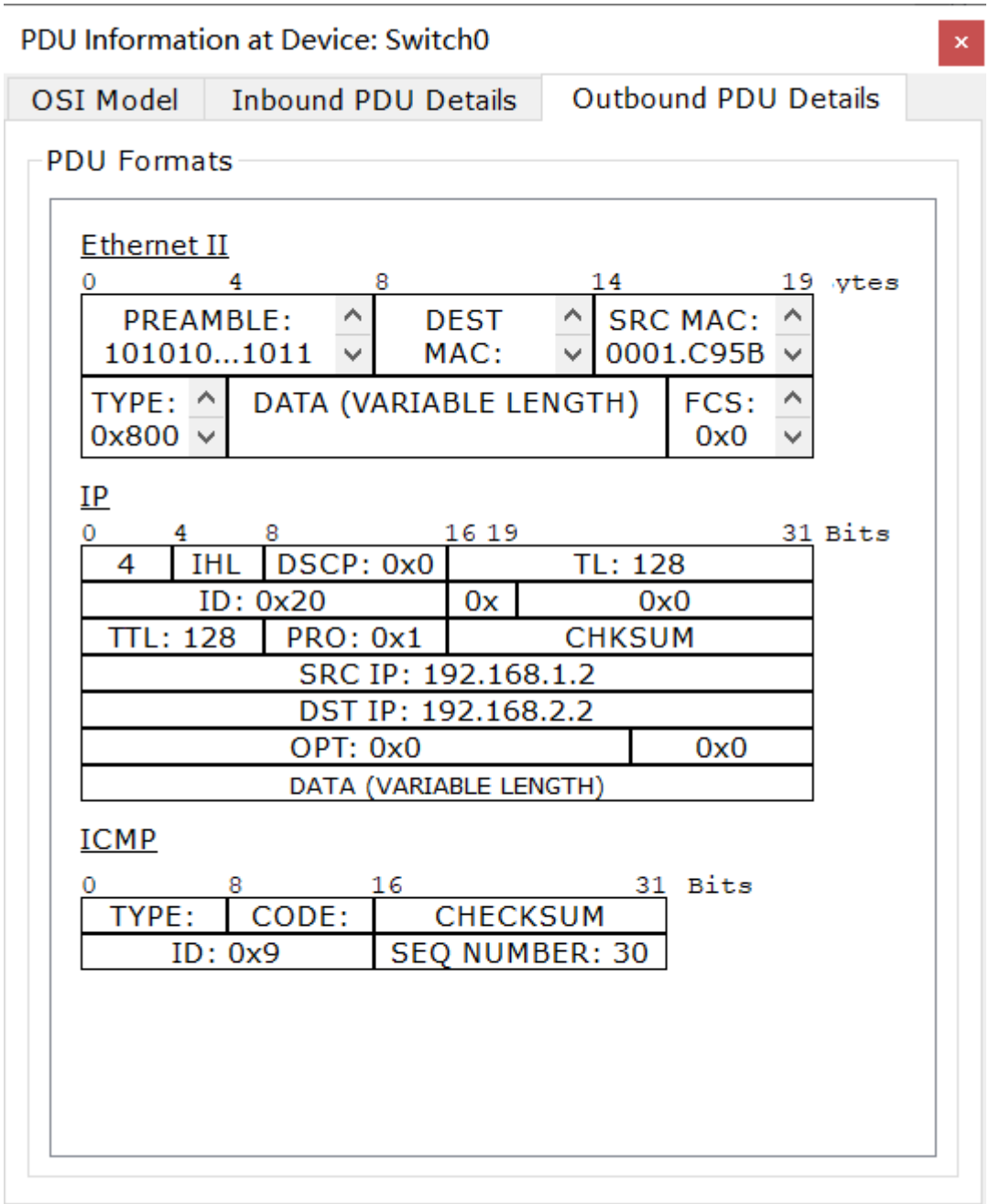



| Vis. | Time(sec) | Last Dev | At Devi | Type | Info |
|---|---|---|---|---|---|
| | 6.005 | PC0 | Switch0 | ICMP | |
| | 6.006 | Switch0 | Router1 | ICMP | |
| | 6.007 | Router1 | Router4 | ICMP | |
| | 6.008 | Router4 | Router5 | ICMP | |
| | 6.009 | Router5 | Router2 | ICMP | |
| | 6.010 | Router2 | Switch1 | ICMP | |
| | 6.011 | Switch1 | Server1 | ICMP | |




| Vis. | Time(sec) | Last Dev | At Devi | Type | Info |
|---|---|---|---|---|---|
| | 6.012 | Server1 | Switch1 | ICMP | |
| | 6.013 | Switch1 | Router2 | ICMP | |
| | 6.014 | Router2 | Router5 | ICMP | |
| | 6.015 | Router5 | Router4 | ICMP | |
| | 6.016 | Router4 | Router1 | ICMP | |
| | 6.017 | Router1 | Switch0 | ICMP | |
| | 6.018 | Switch0 | PC0 | ICMP | |

查看内层 ip 数据报




查看外层 ip 数据报,可以看到，首先被封装成 gre 格式，gre 个石碑封装成外层 ip 分组。之后外层 ip 分组被封装成为 esp 报文，报文作为 esp 报文的净荷，esp 采用的加密算法为 3des，hmac 算法为 hmac-md5，至此实验完毕。

PDU Information at Device: Router4

OSI Model | Inbound PDU Details | Outbound PDU Details

PDU Formats

Ethernet II

0 4 8 14 19 ytes

| | | |
|---|---|---|
| PREAMBLE: 101010...1011 | DEST MAC: | SRC MAC: 00E0.8F62 |
| TYPE: 0x800 | DATA (VARIABLE LENGTH) | FCS: 0x0 |

IP

0 4 8 16 19 31 Bits

| | | | | |
|---|---|---|---|---|
| 4 | IHL | DSCP: 0x0 | TL: 20 | |
| ID: 0xc9b | | | 0x | 0x0 |
| TTL: 254 | | PRO: 0x32 | CHKSUM | |
| SRC IP: 192.1.1.1 | | | | |
| DST IP: 192.1.2.1 | | | | |
| OPT: 0x0 | | | | 0x0 |
| DATA (VARIABLE LENGTH) | | | | |

ENCAPSULATING SECURITY PAYLOA

0 8 16 31 Bits

| |
|---|
| ESP SPI: 1963359709 |
| ESP SEQUENCE: 272 |
| ESP DATA ENCRYPTED WITH 3DES |
| ESP DATA AUTHENTICATED WITH MD5 |

IP

PDU Information at Device: Router4

OSI Model | Inbound PDU Details | Outbound PDU Details

PDU Formats

IP

| 0 | 4 | 8 | 16 19 | 31 Bits |
|---|---|---|---|---|
| 4 | IHL | DSCP: 0x0 | TL: 20 | |
| ID: 0xc9a | | | 0x | 0x0 |
| TTL: 255 | | PRO: 0x2f | CHKSUM | |
| SRC IP: 192.1.1.1 | | | | |
| DST IP: 192.1.2.1 | | | | |
| OPT: 0x0 | | | | 0x0 |
| DATA (VARIABLE LENGTH) | | | | |

GRE

| 0 | 8 | 16 | 31 Bits |
|---|---|---|---|
| FLAGS: 0 | | PROTOCOL TYPE: 0x800 | |

IP

| 0 | 4 | 8 | 16 19 | 31 Bits |
|---|---|---|---|---|
| 4 | IHL | DSCP: 0x0 | TL: 128 | |
| ID: 0x20 | | | 0x | 0x0 |
| TTL: 127 | | PRO: 0x1 | CHKSUM | |
| SRC IP: 192.168.1.2 | | | | |
| DST IP: 192.168.2.2 | | | | |
| OPT: 0x0 | | | | 0x0 |
| DATA (VARIABLE LENGTH) | | | | |

ICMP

| 0 | 8 | 16 | 31 Bits |
|---|---|---|---|
| TYPE: | CODE: | CHECKSUM | |
| ID: 0x9 | | SEQ NUMBER: 30 | |